Fiscal Year:	FY 2022	**Task Last Updated:**	FY 08/26/2025
PI Name:	Mateus, Jaime Ph.D.		
Project Title:	Medical Data Acquisition and Handling in Commercial Space		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jaime.mateus@spacex.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	
Organization Name:	Space Exploration Technologies		
PI Address 1:	1 Rocket Rd.		
PI Address 2:			
PI Web Page:			
City:	Hawthorne	**State:**	CA
Zip Code:	90250	**Congressional District:**	43
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Catalyst
Start Date:	12/01/2020	**End Date:**	11/30/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Per TRISH, the Principal Investigator (PI) for this project changed from Anil Menon, MD to Jaime Mateus, Ph.D. (Ed., 7/9/25). Dr. Menon remained as a Co-Investigator on the project.		
COI Name (Institution):	Menon, Anil (Space Exploration Technologies)		
Grant/Contract No.:	NNX16AO69A-CAT0002		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Catalyst Project SpaceX is focused on developing the next vehicle to reach the moon and also fly a broader group of passengers to space with more baseline medical problems. In order to achieve this ambitious objective, a data driven approach will be necessary to understand in-flight disease. This grant will help to capture data in-flight, concentrate that data, and permit collaboration with researchers.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	This project was designed to capture data in-flight, concentrate that data, and build a framework for collaboration with researchers. In doing so, the entire field of space medicine will change from a field that depends on case reports to a field that can truly embrace evidence based medicine. An incremental approach was initiated to perform COVID-19 research, publish that data from a commercial space company, and then begin focusing on the human system to engender a culture of scientific openness. Original objectives were to: 1. Analyze and publish data from SARS-CoV-2 testing. a. Assess the risk of SARS-CoV-2 infection in SpaceX employees stratified by exposure risks b. Determine the range and severity of disease due to SARS-CoV-2 infection, including asymptomatic infections c. Inform SpaceX back to work policies d. Characterize immunological and demographic features of asymptomatic infections and presence of reinfections e. Build an internal framework for data collection and data privacy f. Build an interface for data visualization and analysis g. Submit findings as original research to a peer reviewed journal 2. Develop a critical design review and architecture for end-to-end medical data collection from the human, transfer to the ground, storage, and analysis for and in-flight healthcare. a. Scope the medical system that is needed for Starship, including all components b. Develop a trade for sensors that would provide valuable telemetry c. Determine which data needs real-time telemetry and which data can be stored for review later d. Integrate data collection and telemetry into Starship architecture e. Propose a sentinel piece of telemetry that can be integrated into crew Dragon and propose a process for integration f. Develop privacy constraints for data storage g. Propose a model for including medical researchers h. Complete a critical design review that includes all aspects of a medical system and present this design review to SpaceX stakeholders i. Iterate towards a final, acceptable path forward Key findings and impact were 4 published papers on COVID-19, including 1 paper in Nature and Wall Street Journal coverage. Five more papers submitted to top-tier journals; launch and execution of 7 experiments on Inspiration4 with only 4 months of development time, including certification, training, operational products, and execution. Developed openness at commercial space company with regards to sharing of science, publication of medical science, and the importance of the human system and research.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
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